需求管理系统 / 完整字段级 ER 模型

13 个实体 · 151 个字段 · 34 个真实外键 · 主键 / 外键 / 唯一键 / 可空性

身份与权限 **需求与分类** **版本、评审与协作** **受控变更与审计**

User

PK	userId	BIGINT
U1	username	VARCHAR(64)
U2	email	VARCHAR(254)
	passwordHash	VARCHAR(255)
	displayName	VARCHAR(100)
	accountStatus	VARCHAR(32)
	createdAt	DATETIME(6)
	updatedAt	DATETIME(6)
	lockVersion	BIGINT

Requirement

PK/FK	requirementId	BIGINT
U1	requirementKey	VARCHAR(32)
	title	VARCHAR(200)
	description	TEXT
	level	VARCHAR(32)
	kind	VARCHAR(32)
	priority	VARCHAR(32)
	source ?	TEXT
	rationale ?	TEXT
	acceptanceCriteria ?	TEXT
	status	VARCHAR(32)
FK	creatorId	BIGINT
FK	assigneeId ?	BIGINT
FK	currentVersionId ?	BIGINT
	firstSubmittedAt ?	DATETIME(6)
	isWithdrawn	TINYINT
FK	withdrawnBy ?	BIGINT
	withdrawnAt ?	DATETIME(6)
	createdAt	DATETIME(6)
	updatedAt	DATETIME(6)
	lockVersion	BIGINT

RequirementVersion

PK/U2	versionId	BIGINT
FK/U1/U2	requirementId	BIGINT
U1	versionNo	INT
	title	VARCHAR(200)
	description	TEXT
	level	VARCHAR(32)
	kind	VARCHAR(32)
	priority	VARCHAR(32)
	source	TEXT
	rationale	TEXT
	acceptanceCriteria	TEXT
FK/U3	initialReviewId ?	BIGINT
FK/U4	appliedChangeRequestId ?	BIGINT
FK	createdBy	BIGINT
	createdAt	DATETIME(6)
	changeReason	TEXT

ChangeRequest

PK/U1	changeRequestId	BIGINT
FK/U1	requirementId	BIGINT
FK	baseVersionId	BIGINT
	requestTitle	VARCHAR(200)
	reason	TEXT
	proposedTitle	VARCHAR(200)
	proposedDescription	TEXT
	proposedLevel	VARCHAR(32)
	proposedKind	VARCHAR(32)
	proposedPriority	VARCHAR(32)
	proposedSource ?	TEXT
	proposedRationale ?	TEXT
	proposedAcceptanceCriteria ?	TEXT
	status	VARCHAR(32)
FK	createdBy	BIGINT
	createdAt	DATETIME(6)
	updatedAt	DATETIME(6)
FK	appliedBy ?	BIGINT
	appliedAt ?	DATETIME(6)
	lockVersion	BIGINT
U2/G	activeRequirementId ?	BIGINT

UserRole

PK/FK	userId	BIGINT
PK/FK	roleId	BIGINT
FK	grantedBy	BIGINT
	grantedAt	DATETIME(6)

Role

PK	roleId	BIGINT
U1	roleCode	VARCHAR(32)
	roleName	VARCHAR(100)
	description	VARCHAR(500)

RequirementTag

PK/FK	requirementId	BIGINT
PK/FK	tagId	BIGINT
FK	addedBy	BIGINT
	addedAt	DATETIME(6)

Tag

PK	tagId	BIGINT
U1	name	VARCHAR(50)
	description ?	VARCHAR(500)
FK	createdBy	BIGINT
	createdAt	DATETIME(6)

RequirementReview

PK/U2	reviewId	BIGINT
FK/U1/U2	requirementId	BIGINT
U1	roundNo	INT
	snapshotTitle	VARCHAR(200)
	snapshotDescription	TEXT
	snapshotLevel	VARCHAR(32)
	snapshotKind	VARCHAR(32)
	snapshotPriority	VARCHAR(32)
	snapshotSource	TEXT
	snapshotRationale	TEXT
	snapshotAcceptanceCriteria	TEXT
FK	submittedBy	BIGINT
	submittedAt	DATETIME(6)
	reviewStatus	VARCHAR(32)
FK	reviewerId ?	BIGINT
	decision ?	VARCHAR(32)
	comment ?	TEXT
	decidedAt ?	DATETIME(6)
U3/G	activeRequirementId ?	BIGINT

ChangeRequestReview

PK	changeReviewId	BIGINT
FK	requirementId	BIGINT
FK/U1	changeRequestId	BIGINT
U1	roundNo	INT
FK	snapshotBaseVersionId	BIGINT
	snapshotRequestTitle	VARCHAR(200)
	snapshotReason	TEXT
	snapshotProposedTitle	VARCHAR(200)
	snapshotProposedDescription	TEXT
	snapshotProposedLevel	VARCHAR(32)
	snapshotProposedKind	VARCHAR(32)
	snapshotProposedPriority	VARCHAR(32)
	snapshotProposedSource	TEXT
	snapshotProposedRationale	TEXT
	snapshotProposedAcceptanceCriteria	TEXT
FK	submittedBy	BIGINT
	submittedAt	DATETIME(6)
	reviewStatus	VARCHAR(32)
FK	reviewerId ?	BIGINT
	decision ?	VARCHAR(32)
	comment ?	TEXT
	decidedAt ?	DATETIME(6)
U2/G	activeChangeRequestId ?	BIGINT

RequirementRelation

PK	relationId	BIGINT
FK/U1	sourceRequirementId	BIGINT
FK/U1	targetRequirementId	BIGINT
U1	relationType	VARCHAR(32)
	description ?	VARCHAR(1000)
FK	createdBy	BIGINT
	createdAt	DATETIME(6)

Comment

PK	commentId	BIGINT
FK	requirementId	BIGINT
FK	authorId	BIGINT
	content	TEXT
	createdAt	DATETIME(6)
	isDeleted	TINYINT
FK	deletedBy ?	BIGINT
	deletedAt ?	DATETIME(6)

AuditEvent

PK	auditId	BIGINT
FK	actorId	BIGINT
FK	requirementId ?	BIGINT
	targetType	VARCHAR(32)
	targetId	VARCHAR(128)
	action	VARCHAR(64)
	outcome	VARCHAR(32)
	beforeData ?	JSON
	afterData ?	JSON
	occurredAt	DATETIME(6)

图例

PK 主键
FK 外键
U1、U2… 实体内的唯一键组
G 数据库生成的约束字段
? 字段可空

实线 标识关系
虚线 非标识关系
○ 0 | 1 三叉 多个

相同 U 编号的字段组成复合唯一键。
R 编号对应 foreign-keys.csv。

审计 targetType / targetId 为逻辑定位，不是外键。完整八项内容、生成列及同需求复合引用均保留。